Milestone 1: Data Acquisition, Cleaning & Initial Exploration

1 Import Libraries

In [4]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

2 Load Dataset

In [5]:
```
df = pd.read_csv("us_disaster_declarations.csv")
df.head()
```

Out[5]:

| | fema_declaration_string | disaster_number | state | declaration_type | declaration_date | fy_d |
|---|---|---|---|---|---|---|
| **0** | DR-1-GA | 1 | GA | DR | 1953-05-02T00:00:00Z | |
| **1** | DR-2-TX | 2 | TX | DR | 1953-05-15T00:00:00Z | |
| **2** | DR-3-LA | 3 | LA | DR | 1953-05-29T00:00:00Z | |
| **3** | DR-4-MI | 4 | MI | DR | 1953-06-02T00:00:00Z | |
| **4** | DR-5-MT | 5 | MT | DR | 1953-06-06T00:00:00Z | |

5 rows × 27 columns

3 Inspect Dataset

In [6]:
```
df.info()
df.describe()
df.columns
```

```
<class 'pandas.DataFrame'>
RangeIndex: 68485 entries, 0 to 68484
Data columns (total 27 columns):
 #   Column                      Non-Null Count  Dtype
---  ------                      --------------  -----
 0   fema_declaration_string     68485 non-null  str
 1   disaster_number             68485 non-null  int64
 2   state                       68485 non-null  str
 3   declaration_type            68485 non-null  str
 4   declaration_date            68485 non-null  str
 5   fy_declared                 68485 non-null  int64
 6   incident_type               68485 non-null  str
 7   declaration_title           68485 non-null  str
 8   ih_program_declared         68485 non-null  int64
 9   ia_program_declared         68485 non-null  int64
 10  pa_program_declared         68485 non-null  int64
 11  hm_program_declared         68485 non-null  int64
 12  incident_begin_date         68485 non-null  str
 13  incident_end_date           67963 non-null  str
 14  disaster_closeout_date      52068 non-null  str
 15  tribal_request              68485 non-null  int64
 16  fips                        68485 non-null  int64
 17  place_code                  68485 non-null  int64
 18  designated_area             68485 non-null  str
 19  declaration_request_number  68485 non-null  int64
 20  last_ia_filing_date         19395 non-null  str
 21  incident_id                 68485 non-null  int64
 22  region                      68485 non-null  int64
 23  designated_incident_types   20673 non-null  str
 24  last_refresh                68485 non-null  str
 25  hash                        68485 non-null  str
 26  id                          68485 non-null  str
dtypes: int64(12), str(15)
memory usage: 14.1 MB
```

Out[6]:

```
Index(['fema_declaration_string', 'disaster_number', 'state',
       'declaration_type', 'declaration_date', 'fy_declared', 'incident_type',
       'declaration_title', 'ih_program_declared', 'ia_program_declared',
       'pa_program_declared', 'hm_program_declared', 'incident_begin_date',
       'incident_end_date', 'disaster_closeout_date', 'tribal_request', 'fips',
       'place_code', 'designated_area', 'declaration_request_number',
       'last_ia_filing_date', 'incident_id', 'region',
       'designated_incident_types', 'last_refresh', 'hash', 'id'],
      dtype='str')
```

## 4 Check Missing Values

In [12]:

```
df.isnull().sum()
```

Out[12]:
```
fema_declaration_string       0
disaster_number               0
state                         0
declaration_type              0
declaration_date              0
fy_declared                   0
incident_type                 0
declaration_title             0
ih_program_declared           0
ia_program_declared           0
pa_program_declared           0
hm_program_declared           0
incident_begin_date           0
tribal_request                0
fips                          0
place_code                    0
designated_area               0
declaration_request_number    0
incident_id                   0
region                        0
last_refresh                  0
hash                          0
id                            0
dtype: int64
```

5 Handle Missing Values

In [16]:
```
df = df.dropna(subset=['declaration_date'])
```

In [17]:
```
df
```

Out[17]:

| | fema_declaration_string | disaster_number | state | declaration_type | declaration_date |
|---|---|---|---|---|---|
| **0** | DR-1-GA | 1 | GA | DR | 1953-05-02T00:00:00Z |
| **1** | DR-2-TX | 2 | TX | DR | 1953-05-15T00:00:00Z |
| **2** | DR-3-LA | 3 | LA | DR | 1953-05-29T00:00:00Z |
| **3** | DR-4-MI | 4 | MI | DR | 1953-06-02T00:00:00Z |
| **4** | DR-5-MT | 5 | MT | DR | 1953-06-06T00:00:00Z |
| **...** | ... | ... | ... | ... | ... |
| **68480** | DR-4892-WI | 4892 | WI | DR | 2025-09-11T00:00:00Z |
| **68481** | DR-4892-WI | 4892 | WI | DR | 2025-09-11T00:00:00Z |
| **68482** | DR-4890-SD | 4890 | SD | DR | 2025-09-11T00:00:00Z |
| **68483** | FM-5613-HI | 5613 | HI | FM | 2025-09-23T00:00:00Z |
| **68484** | FM-5614-WA | 5614 | WA | FM | 2025-09-26T00:00:00Z |

68485 rows × 23 columns

## 6 Convert Date Columns

In [19]:
```
df['declaration_date'] = pd.to_datetime(df['declaration_date'])
df
```

Out[19]:

| | fema_declaration_string | disaster_number | state | declaration_type | declaration_date |
|---|---|---|---|---|---|
| **0** | DR-1-GA | 1 | GA | DR | 1953-05-02 00:00:00+00:00 |
| **1** | DR-2-TX | 2 | TX | DR | 1953-05-15 00:00:00+00:00 |
| **2** | DR-3-LA | 3 | LA | DR | 1953-05-29 00:00:00+00:00 |
| **3** | DR-4-MI | 4 | MI | DR | 1953-06-02 00:00:00+00:00 |
| **4** | DR-5-MT | 5 | MT | DR | 1953-06-06 00:00:00+00:00 |
| **...** | ... | ... | ... | ... | ... |
| **68480** | DR-4892-WI | 4892 | WI | DR | 2025-09-11 00:00:00+00:00 |
| **68481** | DR-4892-WI | 4892 | WI | DR | 2025-09-11 00:00:00+00:00 |
| **68482** | DR-4890-SD | 4890 | SD | DR | 2025-09-11 00:00:00+00:00 |
| **68483** | FM-5613-HI | 5613 | HI | FM | 2025-09-23 00:00:00+00:00 |
| **68484** | FM-5614-WA | 5614 | WA | FM | 2025-09-26 00:00:00+00:00 |

68485 rows × 23 columns

## 7 Extract Year

In [23]:
```
df['year'] = df['declaration_date'].dt.year
df
```

Out[23]:

| | fema_declaration_string | disaster_number | state | declaration_type | declaration_date |
|---|---|---|---|---|---|
| **0** | DR-1-GA | 1 | GA | DR | 1953-05-02 00:00:00+00:00 |
| **1** | DR-2-TX | 2 | TX | DR | 1953-05-15 00:00:00+00:00 |
| **2** | DR-3-LA | 3 | LA | DR | 1953-05-29 00:00:00+00:00 |
| **3** | DR-4-MI | 4 | MI | DR | 1953-06-02 00:00:00+00:00 |
| **4** | DR-5-MT | 5 | MT | DR | 1953-06-06 00:00:00+00:00 |
| **...** | ... | ... | ... | ... | ... |
| **68480** | DR-4892-WI | 4892 | WI | DR | 2025-09-11 00:00:00+00:00 |
| **68481** | DR-4892-WI | 4892 | WI | DR | 2025-09-11 00:00:00+00:00 |
| **68482** | DR-4890-SD | 4890 | SD | DR | 2025-09-11 00:00:00+00:00 |
| **68483** | FM-5613-HI | 5613 | HI | FM | 2025-09-23 00:00:00+00:00 |
| **68484** | FM-5614-WA | 5614 | WA | FM | 2025-09-26 00:00:00+00:00 |

68485 rows × 24 columns

## 8 Basic Frequency Counts

In [24]:
```
df['state'].value_counts().head()
df
```

Out[24]:

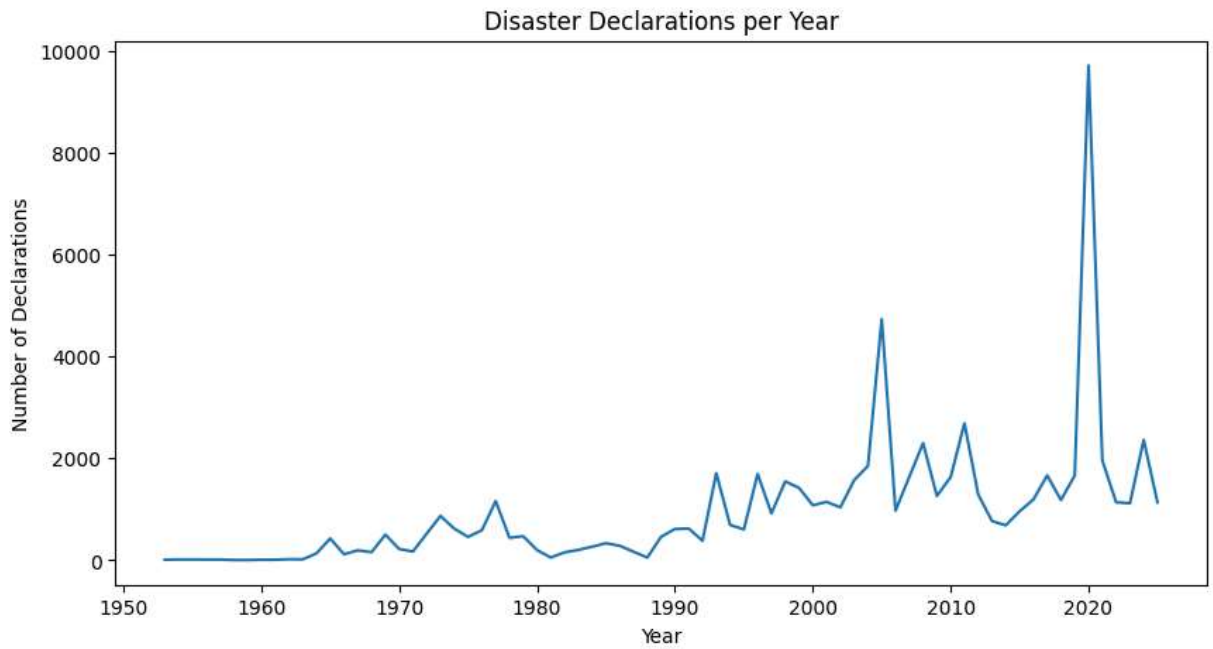
| | fema_declaration_string | disaster_number | state | declaration_type | declaration_date |
|---|---|---|---|---|---|
| **0** | DR-1-GA | 1 | GA | DR | 1953-05-02 00:00:00+00:00 |
| **1** | DR-2-TX | 2 | TX | DR | 1953-05-15 00:00:00+00:00 |
| **2** | DR-3-LA | 3 | LA | DR | 1953-05-29 00:00:00+00:00 |
| **3** | DR-4-MI | 4 | MI | DR | 1953-06-02 00:00:00+00:00 |
| **4** | DR-5-MT | 5 | MT | DR | 1953-06-06 00:00:00+00:00 |
| **...** | ... | ... | ... | ... | ... |
| **68480** | DR-4892-WI | 4892 | WI | DR | 2025-09-11 00:00:00+00:00 |
| **68481** | DR-4892-WI | 4892 | WI | DR | 2025-09-11 00:00:00+00:00 |
| **68482** | DR-4890-SD | 4890 | SD | DR | 2025-09-11 00:00:00+00:00 |
| **68483** | FM-5613-HI | 5613 | HI | FM | 2025-09-23 00:00:00+00:00 |
| **68484** | FM-5614-WA | 5614 | WA | FM | 2025-09-26 00:00:00+00:00 |

68485 rows × 24 columns

### 9 Declarations per Year

In [25]:
```
yearly_counts = df.groupby('year').size()

plt.figure(figsize=(10,5))
yearly_counts.plot()
plt.title("Disaster Declarations per Year")
plt.xlabel("Year")
plt.ylabel("Number of Declarations")
plt.show()
```

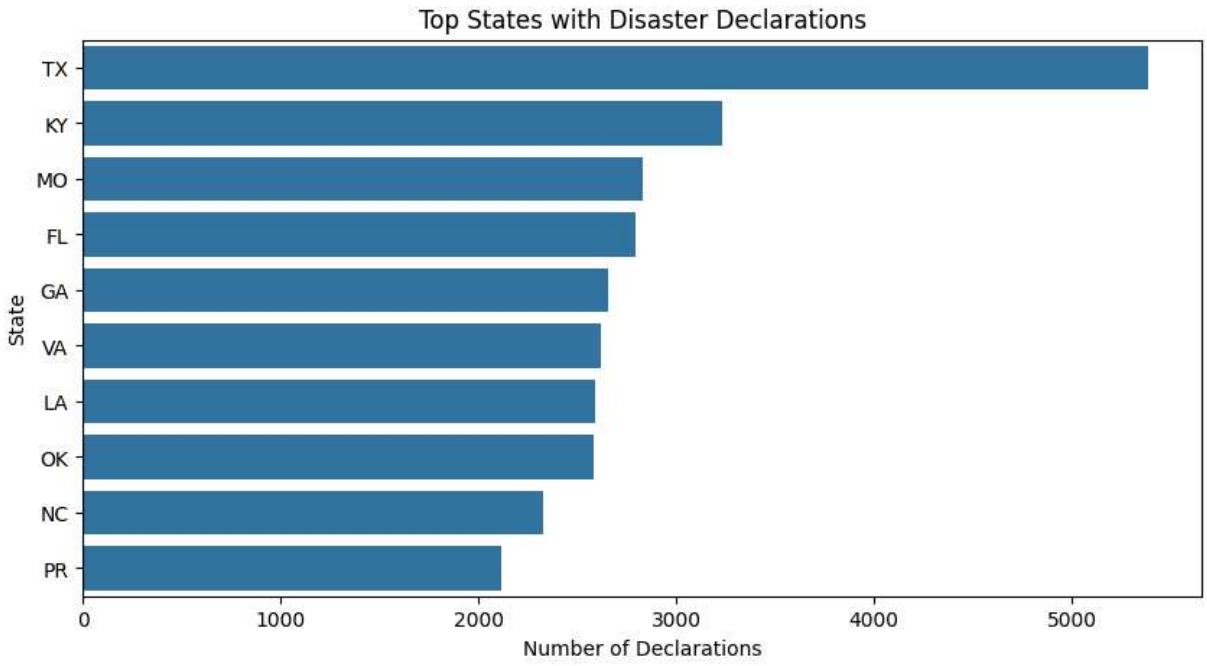



### 10 Declarations per State

In [26]:
```
import seaborn as sns
state_counts = df['state'].value_counts().head(10)

plt.figure(figsize=(10,5))
sns.barplot(x=state_counts.values, y=state_counts.index)

plt.title("Top States with Disaster Declarations")
plt.xlabel("Number of Declarations")
plt.ylabel("State")
plt.show()
```




In Milestone-1, the FEMA disaster dataset was loaded and cleaned by handling missing values and formatting date columns. Initial exploratory analysis revealed patterns in disaster declarations across years and states. Preliminary visualizations highlighted increasing disaster frequency and state-level differences, providing a foundation for deeper analysis in later milestones.